Uncovering Mechanobiological Drivers of Foam Cell Formation via Integrated Agent-Based and Explainable Machine Learning

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Introduction

Atherosclerosis (AS) is influenced by inflammation and biomechanics and characterized by foam cell (FC) formation. It has no cure beyond a very early stage. Thus, new approaches are necessary for subclinical characterization. We aimed to create a model of early AS, including key atherogenic processes to support longitudinal risk stratification and therapeutic target identification.

Methods

A regulatory network model (RNM) was combined with an agent-based model (ABM). The RNM was solved with Boolean-based differential equations [1]. It interacted with higher-level agent rules integrating oxidative stress; inflammation; LDL oxidation & aggregation; wall shear stress (WSS); reverse cholesterol transport (RCT). The integrated model was implemented in BioDynaMo. 18 cases were tested with low, borderline, and normal HDL levels (25; 50; 75 mg/dL) combined with normal and low WSS. Each combination was simulated with high, borderline, and normal LDL: 200; 150; 90 mg/dL. Each simulation was repeated 10 times. A sensitivity analysis explored the effects of the rates of: LDL uptake by CD36, SR-A1 & LRP1 receptors; RCT by ABCA1 transporter; LDL modification. Four ensemble models (Random Forest, Gradient Boosting, XGBoost, LightGBM) were trained on 42 model features over the 576 simulation outcomes, to identify the most pro-atherogenic and protective factors. Permutation importance analysis (PIA) was conducted on the best performing model with 100-repeat shuffling and one-sample t-test. A pharmacodynamics module with statin perturbation validated the model against a real patient cohort from Singh et al. [2].

Results

The gradient boosting model performed best (R^2=1.000, MAE=3.99 FC). PIA revealed no single factor of importance >1.0 indicating that the model captured the multi-factorial nature of AS. At year 1, the protective anti-inflammatory expression of TGF-β was the main determinant of FC emergence. At year 2, macrophage (MC) inflammation (transcription factor NF-κB & cytokine IFN-γ) mainly drove FC. Endothelial (EC) superoxide production mainly predicted AS at year 3. High WSS showed a protective role followed by atherogenic MC-CD36 and accumulation of dead MC (Fig. 1).

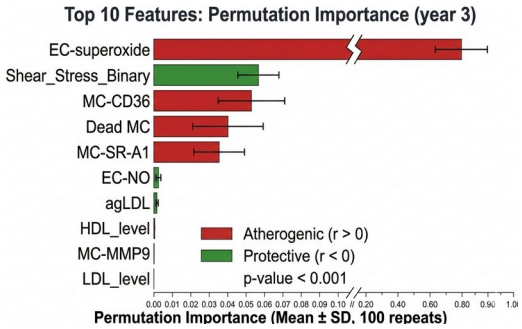


Fig. 1: The main year 3 AS predictors based on PIA. NO: nitric oxide, agLDL: aggregated LDL, MMP9: matrix metalloproteinase 9.

Validation study showed the model ability to replicate statin effects on LDL (AS model median=35% vs Singh et al. 33.3%) and plaque oxidized LDL reductions (AS model median=26.9% vs. Singh et al. 28%) (Fig. 2).

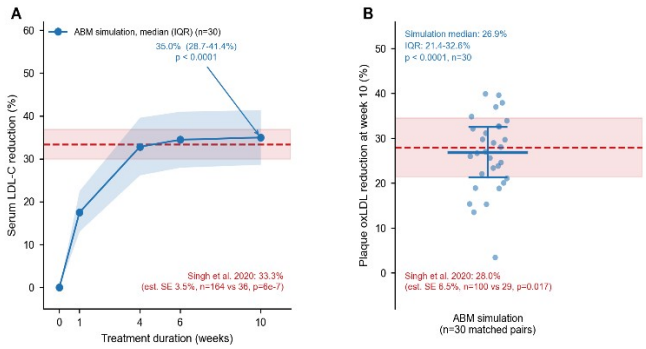


Fig. 2: AS Model validation against the Singh et al. patient cohort.

Conclusions

The hybrid AS model uniquely reveals the temporal evolution of FC drivers, accounting for the diversity of lipid profiles. It shows that AS is caused by combinations of several conditions along time. Anti-inflammatory signalling appeared protective, at first, but lost influence at year 3, where EC superoxide had become a key driver of FC, which is in line with literature. Some patients develop CVD with normal LDL [3]. The model suggests that anti-inflammatory approaches could be effective in early risk mitigation. It further reveals the role of dead MC in increasing FC (Fig. 1), pointing at an underexplored therapeutic target - MC efferocytosis. Enhanced clearance of dead MC could break the vicious cycle of inflammation and cell death, preventing AS progression.

Acknowledgements

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References

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3. Parton et al., J Brief. Bioinform; 20:932,